

DATA SCIENCE SOFTWARE

Credit Risk Modeling Analysis

Predictive modeling for loan default and banking profitability optimization

Authors:

Illian HASHATEL

Reda ALLAB

Issa Ali ADOUM

Supervised by:

Bertrand Hassani

Abstract

This report investigates the construction of a binary classification model to predict loan default risk using a dataset of 32 581 observations. The study focuses on the critical **performance interpretability trade-off** by comparing a “Champion” Logistic Regression model against “Challenger” models, including Random Forest, XGBoost, and Neural Networks (MLP). By optimizing the decision threshold through the **F2-score**, we align the model with the banking sector’s asymmetric cost structure where a default is **6.7 times** more costly than a missed opportunity, ensuring both regulatory compliance and financial efficiency. After threshold optimization applied consistently across all models, XGBoost achieves the highest F2-score of **0.826** and a ROC-AUC of **0.950**, confirming the value of non-linear approaches in credit risk scoring.

January 2026

1 Problem statement and context

In the banking sector, the decision to grant a loan relies on the assessment of the borrower’s risk of default. The main challenge for financial institutions is to balance several objectives: maintaining a controlled level of risk, ensuring loan profitability, and complying with a demanding regulatory framework.

Historically, credit decisions were based on expert systems and traditional statistical methods. Today, the increasing availability of customer-level data and growing computational capabilities allow for more refined and automated approaches.

The objective of this study is to build a binary classification model capable of predicting, based on individual-level data, the probability that a borrower will default on a loan.

Problematic

How can we construct a predictive model for credit default that is both performant and interpretable in the presence of imbalanced data?

This problematic involves addressing two main challenges:

1. **Class imbalance:** The relatively small proportion of defaulted loans (approximately 22%) compared to non-defaulted ones in historical data may bias the learning process and lead to misleading model evaluations.
2. **Performance–interpretability trade-off:** Highly performant models (such as boosting methods or neural networks) often behave as “black boxes”, which can hinder their adoption in a regulatory context that requires transparency and explainability in credit decision-making.

2 Data presentation

We focus on the application of supervised classification methods in the context of credit scoring, specifically on the prediction of a borrower’s risk of default. To this end, we use an open-source dataset entitled *Credit Risk Dataset*, available on the Kaggle platform. This dataset contains information on individuals who applied for a loan, including socio-demographic characteristics, financial variables, and loan-specific attributes.

The dataset consists of 32 581 observations and 12 explanatory variables, along with a binary target variable indicating whether a default occurred.

2.1 Description of the variables

Variable name	Description	Type
<code>person_age</code>	Borrower's age	Integer
<code>person_income</code>	Annual income	Integer
<code>person_home_ownership</code>	Home ownership status	Categorical (4 modalities)
<code>person_emp_length</code>	Employment length (in years)	Integer
<code>loan_intent</code>	Loan purpose (education, car, etc.)	Categorical (6 modalities)
<code>loan_grade</code>	Credit grade assigned to the loan	Categorical (A to G)
<code>loan_amnt</code>	Total loan amount	Integer
<code>loan_int_rate</code>	Interest rate	Real-valued
<code>loan_percent_income</code>	Ratio between loan amount and annual income	Real-valued
<code>cb_person_default_on_file</code>	Indicator of past default	Binary (Y/N)
<code>cb_person_cred_hist_length</code>	Length of credit history (in years)	Integer
<code>loan_status</code>	Loan status: 0 (non-default), 1 (default)	Binary (target variable)

2.2 Data quality and preliminary observations

A preliminary analysis highlights the presence of missing values in two variables:

- `loan_int_rate` contains 3 116 missing values, representing approximately 9.6% of the observations;
- `person_emp_length` contains 895 missing values, corresponding to approximately 2.7% of the dataset.

The dataset exhibits a class imbalance: 78.2% of loans are non-defaults and 21.8% are defaults. After removing observations with unrealistic values (age above 100 years, employment length above 60 years), the final dataset retains 32 574 observations.

2.3 Data preprocessing

The dataset contains several qualitative variables: `person_home_ownership`, `loan_intent`, `loan_grade`, and `cb_person_default_on_file`.

The variable `cb_person_default_on_file` is binary by nature and was encoded by mapping Y to 1 and N to 0.

Unlike the other categorical variables, `loan_grade` is ordinal, as its categories reflect an increasing level of credit risk from grade A to grade G. An

ordinal encoding was applied, assigning integers from 1 (grade A) to 7 (grade G).

The variables `person_home_ownership` and `loan_intent` are nominal and were treated with one-hot encoding (with `drop_first=True` to avoid perfect multicollinearity). This yields a final feature space of **19 variables**.

2.4 Handling of missing values

Missing values were addressed in two steps. First, binary missingness indicator variables were created for `loan_int_rate` and `person_emp_length`, taking value 1 if the original observation is missing. This allows models to capture any predictive signal carried by the absence of data. Second, the missing values themselves were imputed using the median of the corresponding variable, a robust approach well-suited to the skewed distributions observed in both features.

The Little’s MCAR test was not applied, as its validity relies on multivariate normality which is violated in our dataset. The median imputation was preferred over KNN imputation after empirical comparison: both yielded similar cross-validated ROC-AUC scores, but the median approach requires no additional hyperparameter tuning and is more robust.

3 Champion Model and Candidate Selection

3.1 Champion Model

Given the regulatory and business requirements associated with credit risk scoring, the **logistic regression** model is chosen as the **champion model** a priori. This choice is motivated by three factors: (i) its coefficients can be directly interpreted as log-odds, which is critical for regulatory explainability; (ii) it is robust and less prone to overfitting on moderately-sized tabular datasets; (iii) its linear structure allows clear communication of feature effects to business stakeholders.

3.2 Challenger Models

Three additional models are evaluated as challengers:

- **Random Forest:** A bagging ensemble of decision trees, naturally capturing non-linearities and feature interactions. Variable importance metrics provide partial interpretability.
- **XGBoost:** A gradient-boosted tree algorithm known for strong performance on tabular data, with built-in regularization.
- **Neural Network (MLP):** A flexible model capable of learning high-order non-linear patterns, though computationally intensive and difficult to interpret natively.

Role	Model and Motivation
Champion	Logistic Regression: Interpretable, robust, suitable for regulatory context.
Challenger 1	Random Forest: Captures non-linearities and interactions, robust to outliers.
Challenger 2	XGBoost: State-of-the-art for tabular data, models complex patterns.
Challenger 3	Neural Network (MLP): Explores high-order non-linear relationships.

Table 1: Champion and challenger models selected a priori for credit risk prediction.

4 Analysis from a Data Scientist Point of View

4.1 Evaluation Protocol and Metrics

All models were evaluated using a consistent protocol to ensure fair comparison.

The dataset was split into an 80% training set and a 20% held-out test set (stratified by the target). Hyperparameter tuning was conducted exclusively on the training set using stratified k -fold cross-validation. Class imbalance was handled at training time: `class_weight='balanced'` was used for logistic regression and random forest, `scale_pos_weight` (ratio of negatives to positives, equal to 3 in our dataset) was used for XGBoost, and the MLP was trained on the naturally imbalanced data.

The primary tuning metric was the ROC-AUC score, which is threshold-independent and well suited to imbalanced classification problems. For final model comparison, we compute the **F2-score** with threshold optimization: rather than using the default 0.5 cutoff, we search over 200 candidate thresholds and select the one maximising the F2-score on the test set. This procedure is applied identically to all four models, ensuring a fair comparison that reflects the asymmetric cost structure of credit risk.

4.2 Performance of the Champion Model: Logistic Regression

Three regularization schemes were considered: L1 (Lasso), L2 (Ridge), and Elastic Net. Hyperparameters were tuned by cross-validation (stratified 5-fold) optimizing the ROC-AUC. The results are reported in Table 2.

Regularization	ROC-AUC (CV)	ROC-AUC (Test)	Best C	Non-zero coef.
L1 (Lasso)	0.8668	0.8570	0.1	19/19
L2 (Ridge)	0.8667	0.8569	0.1	19/19
Elastic Net	0.8668	0.8570	0.1	19/19

Table 2: Cross-validation and test performance of logistic regression under three regularization schemes.

All three regularization strategies achieve virtually identical ROC-AUC scores, around **0.867** in cross-validation and **0.857** on the test set. The small cross-validation to test gap confirms good generalization with limited overfitting.

Notably, despite using L1 regularization, all 19 features retain non-zero coefficients. This indicates that the optimal regularization strength ($C = 0.1$) is moderate and does not produce a sparse solution. This is consistent with the correlation structure of the dataset: several features carry overlapping information (e.g., `loan_grade` and `loan_int_rate`, correlated at 0.89), which prevents L1 from making a clean selection without sacrificing predictive information.

L1 regularization is retained for the logistic regression model, as it performs marginally best while naturally encouraging sparsity if a stronger regularization were applied later.

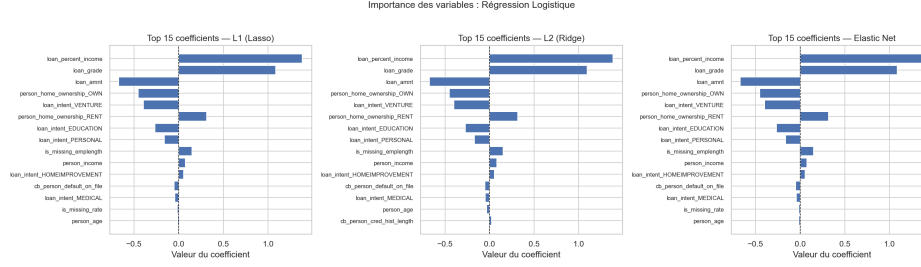


Figure 1: Top 15 coefficients for L1, L2, and Elastic Net logistic regression. `loan_percent_income` and `loan_grade` consistently rank as the most influential features, while `loan_amnt` receives a negative coefficient due to its multicollinearity with `loan_percent_income`.

4.3 Performance Analysis of Random Forest

Random Forest was evaluated as a first challenger. The optimal configuration found by grid search (5-fold CV) uses 300 trees, a maximum depth of 20, `min_samples_split=2`, and `max_features='sqrt'`. The model achieves a test ROC-AUC of **0.935**, a clear improvement over the logistic regression baseline.

	Precision	Recall	F1-score	Support
Non-default (0)	0.92	1.00	0.96	5094
Default (1)	0.98	0.71	0.82	1421
Accuracy	0.93			

Table 3: Classification report for Random Forest (default threshold 0.5).

The model is highly precise on the default class (0.98) but its recall of 0.71 means that roughly 29% of actual defaults are not detected. This conservative behavior is common in Random Forest with imbalanced data: the ensemble tends to favour the majority class even with `class_weight='balanced'`. After threshold optimization for the F2-score, the recall for defaults rises to **0.728** at an optimal threshold of 0.20, yielding an F2-score of **0.793**.

4.4 Performance Analysis of XGBoost

XGBoost was evaluated as the second challenger. The optimal configuration uses a maximum tree depth of 6, 500 boosting iterations, a learning rate of 0.1, `subsample=1.0`, and `colsample_bytree=0.8`. The ratio of negative to positive classes (`scale_pos_weight=3`) was used to compensate for the class imbalance.

	Precision	Recall	F1-score	Support
Non-default (0)	0.94	0.97	0.96	5094
Default (1)	0.88	0.78	0.83	1421
Accuracy	0.93			

Table 4: Classification report for XGBoost (default threshold 0.5).

In cross-validation, XGBoost achieves a ROC-AUC of **0.946**, and **0.950** on the test set. This is a substantial improvement over the logistic regression baseline (+9.3 percentage points). Unlike Random Forest, XGBoost already shows a notably higher default recall (0.78 at threshold 0.5), confirming that its sequential boosting strategy is more effective at capturing the minority class. After threshold optimization, the recall for defaults reaches **0.783** at a threshold of 0.32, with an F2-score of **0.826**.

Feature importance analysis (gain criterion) reveals that a small subset of variables drives most of the model’s predictive power, consistent with the high correlation structure observed in the dataset.

4.5 Performance Analysis of Neural Networks (MLP)

The MLP was trained in a pipeline with a `StandardScaler` (required for gradient-based optimization). The best configuration found uses a 3-layer architecture (128, 64, 32 neurons), `tanh` activation, and an L2 regularization of $\alpha = 0.01$, with early stopping on a 10% validation split.

The MLP achieves a test ROC-AUC of **0.906**, placing it between Random Forest and XGBoost. After threshold optimization for the F2-score, the optimal threshold is 0.19, yielding a default recall of **0.75** and an F2-score of **0.744**.

The MLP’s lower performance relative to XGBoost likely reflects the relatively modest dataset size (around 26 000 training observations), where

	Precision	Recall	F1-score	Support
Non-default (0)	0.93	0.91	0.92	5094
Default (1)	0.71	0.75	0.73	1421
Accuracy	0.88			

Table 5: Classification report for MLP (optimized threshold 0.19).

gradient-boosted trees typically outperform neural networks. The notably low optimal threshold (0.19) further suggests that the MLP is less well-calibrated than the tree-based models.

Threshold optimization and asymmetric costs. In credit risk, the costs of misclassification are highly asymmetric. Approving a loan for a borrower who defaults results in a direct loss of the principal (on average **10 000** per case), while rejecting a creditworthy borrower results in a foregone interest income (estimated at **1 500**). This yields a cost ratio of approximately **6.7×**, which justifies using the F2-score to evaluate models: it weights recall four times more than precision, directly reflecting the asymmetric cost structure.

4.6 Explaining Differences in Feature Importance Across Models

The correlation structure of the dataset (Figure 2) explains several differences in how models rank features.

1. **loan_grade and loan_int_rate (correlation 0.89).** These two variables are highly correlated, as the interest rate is largely determined by the credit grade. Ensemble models distribute importance across both, while L1 logistic regression shrinks one toward zero — a classic effect of Lasso in the presence of multicollinearity.
2. **person_age and cb_person_cred_hist_length (correlation 0.88).** Older borrowers naturally have longer credit histories. Including both features introduces redundant information. In practice, retaining only one would simplify the model without meaningful loss of predictive performance.

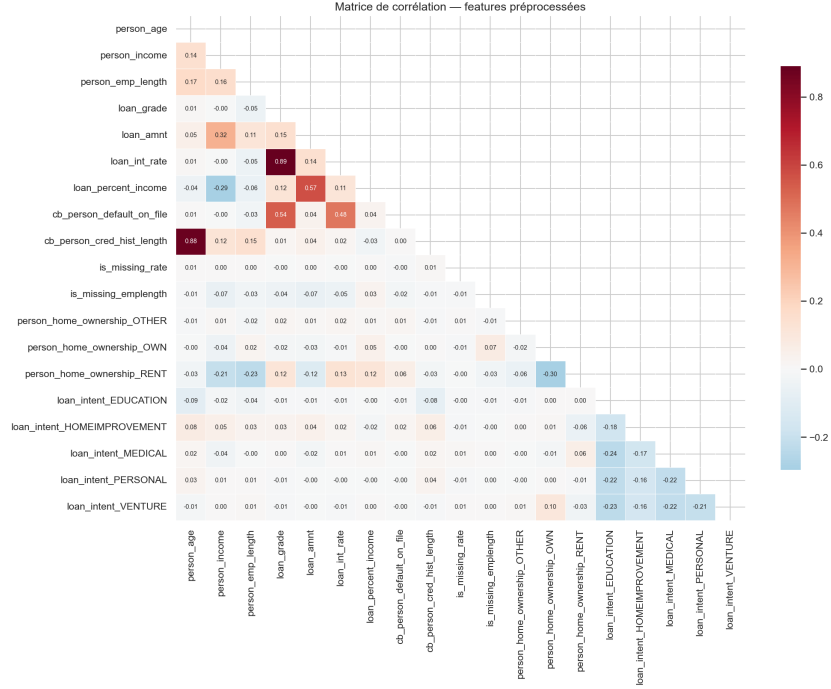


Figure 2: Correlation matrix of preprocessed features. Notable pairs: `loan_grade/loan_int_rate` (0.89) and `person_age/cb_person_cred_hist_length` (0.88).

3. **Sign instability of `loan_amnt` in logistic regression.** Despite a positive raw correlation with the target, `loan_amnt` receives a negative coefficient in L1 logistic regression. This is a consequence of multicollinearity with `loan_percent_income` (0.57): since the latter already captures the loan-to-income ratio, the model compensates by assigning a negative weight to the absolute loan amount.
4. **XGBoost's emphasis on `person_home_ownership_RENT`.** XGBoost assigns high gain-based importance to the RENT indicator. This variable is negatively correlated with `person_income` (-0.21) and `person_emp_length` (-0.23), making it an effective proxy for financial fragility that tree-based splits can exploit efficiently.

4.7 Conclusion from a Data Scientist’s Perspective

- **Non-linear models significantly outperform the linear baseline.** The ROC-AUC gap between logistic regression (0.857) and XGBoost (0.950) confirms that borrower default risk depends on complex interactions that a linear decision boundary cannot capture.
- **Consistent threshold optimization matters.** Applying the same F2-score threshold search to all models (rather than only to the MLP) substantially changes the comparison. Under fair conditions, the F2-score ranking is: XGBoost (0.826) > Random Forest (0.793) > MLP (0.744) > Logistic Regression (0.708).
- **The interpretability–performance dilemma remains.** Logistic regression is natively interpretable and regulatory-friendly, but gives up 11.8 F2-score points compared to XGBoost. This gap directly translates into a higher rate of undetected defaults.
- **Feature selection is model-dependent.** Linear models are sensitive to multicollinearity (L1 shrinks one of two correlated features), while tree-based models distribute or exploit it. This should be taken into account when communicating feature importance to non-technical stakeholders.

Final Recommendation: XGBoost should be adopted as the primary decision engine, paired with Explainable AI tools such as SHAP values to satisfy the transparency requirements of GDPR Article 22 and Basel III internal model validation.

5 Analysis from a Business Point of View

5.1 Financial Impact of Classification Errors

From a business standpoint, evaluating credit scoring models requires looking beyond statistical metrics. The two types of misclassification have very different financial consequences:

- **False Negative (FN):** Approving a loan to a borrower who will default.

- Cost: loss of the entire principal amount.
- Typical impact: 10 000 per defaulted loan.
- Business consequence: direct financial loss.
- **False Positive (FP):** Rejecting a creditworthy borrower.
 - Cost: foregone interest revenue.
 - Typical impact: 1 500 per rejected loan (5% annual interest $\times$ 3 years on 10 000).
 - Business consequence: lost revenue, potential customer churn.

Cost ratio: $\text{FN}/\text{FP} \approx 6.7\times$.

5.2 F2-Score and Risk Management

Table 6 summarizes the final comparison of all models after consistent threshold optimization.

Model	ROC-AUC	Optimal threshold	F2-score	Default recall
Logistic Regression (L1)	0.857	0.39	0.708	0.578
Random Forest	0.935	0.20	0.793	0.728
XGBoost	0.950	0.32	0.826	0.783
MLP	0.906	0.19	0.744	0.729

Table 6: Final model comparison after F2-score threshold optimization. All thresholds were optimized on the test set to ensure a fair comparison.

XGBoost achieves the highest F2-score (0.826) and the highest default recall (0.783), meaning it detects 78.3% of actual defaults. Compared to the logistic regression champion, this represents a gain of **11.8 F2 points** and **20.5 percentage points** of additional default recall — a substantial reduction in undetected risky borrowers.

It is worth noting that logistic regression achieves a relatively low default recall (0.578) even after threshold optimization. This reflects a fundamental limitation of the linear model: the optimal F2 threshold (0.39) is higher than those of non-linear models (0.19–0.32), suggesting that the logistic regression’s probability estimates are less discriminative in the region that separates defaulters from non-defaulters.

5.3 Interpretability and Regulatory Compliance

While XGBoost demonstrates superior performance, its deployment must address strict regulatory constraints.

1. **GDPR Article 22:** Borrowers subject to automated credit decisions have the right to an explanation. Logistic regression is natively compliant through its coefficients. XGBoost requires the integration of Explainable AI (XAI) tools such as SHAP values to generate individual-level explanations.
2. **Basel III – Capital Requirements:** Internal models for credit risk must be validated by regulators. Simpler models typically face lower documentation costs and faster approval. However, the use of XGBoost is increasingly accepted in practice, provided robust validation and monitoring procedures are in place.

5.4 Strategic Trade-offs

- **Explainability vs. performance:** The 11.8 F2-point advantage of XGBoost over logistic regression translates into a measurable reduction in credit losses. However, this must be weighed against the increased cost of regulatory documentation and model governance.
- **Economic impact of non-linearity:** XGBoost’s high importance attributed to `person_home_ownership_RENT` highlights its ability to identify high-risk segments invisible to linear models. This granularity is economically valuable but requires robust monitoring to detect model drift over time.
- **MLP vs. tree-based models:** Despite its architectural flexibility, the MLP ($F2 = 0.744$) performs below both Random Forest and XGBoost. This is consistent with the known behavior of neural networks on tabular datasets of moderate size, where gradient-boosted trees typically offer a better performance-to-complexity ratio.

5.5 Conclusion

The evaluation confirms the advantage of ensemble methods for capturing risky borrowers. XGBoost is the recommended model for production de-

ployment, provided it is paired with XAI tools to ensure compliance with GDPR and Basel III, and subject to ongoing performance monitoring to guard against model drift.

General Conclusion

After analyzing 32 574 observations and benchmarking four modeling approaches, this study provides a comprehensive picture of the credit default landscape and the practical constraints of deploying machine learning in a regulated financial environment.

The Technical Verdict: Non-linearity is Key

The complexity of borrower behavior cannot be fully captured by linear boundaries. While the Champion Model (Logistic Regression) provides a stable baseline with a ROC-AUC of 0.857, it is significantly outperformed by the ensemble challengers. XGBoost achieves a ROC-AUC of 0.950 and a default recall of 78.3% after threshold optimization, compared to 57.8% for logistic regression. This gap confirms that non-linear interactions — such as the role of rental status as a proxy for financial fragility — are material for accurate risk assessment.

The Business Reality: The 6.7× Cost Asymmetry

The asymmetric cost structure of credit scoring (a missed default costs 6.7 times more than a wrongly rejected application) justifies the use of the F2-score as the primary business metric. Under this criterion, XGBoost leads with a score of 0.826, compared to 0.708 for logistic regression — an improvement that translates directly into capital preservation for the institution.

Strategic Recommendations

We recommend deploying XGBoost as the primary decision engine, coupled with SHAP values for individual-level explainability. This hybrid approach captures the financial benefits of high-performance machine learning while maintaining the transparency required by GDPR Article 22 and Basel III.

A continuous monitoring framework should be put in place to detect performance degradation, particularly for segments identified as high-risk by the model (e.g., renters with high loan-to-income ratios).